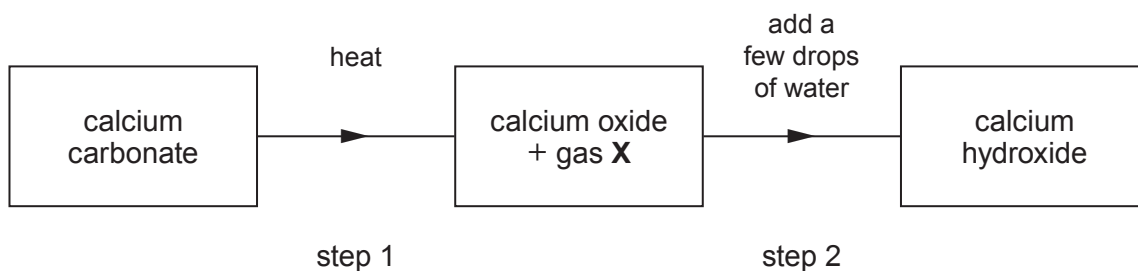


9. (a) The flow diagram shows the reactions carried out to prepare calcium hydroxide from calcium carbonate, CaCO_3 .



- (i) Name the type of reaction taking place in step 1. [1]

.....

- (ii) Give the name of gas X. [1]

.....

- (iii) Give **one** observation that shows that step 2 is exothermic. [1]

.....

- (iv) Write a balanced equation for the reaction of calcium oxide and water to form calcium hydroxide. [2]

.....

- (b) Give a reason why no reaction occurs when sodium carbonate is heated with a Bunsen burner. [1]

.....

.....

- (c) Barium hydroxide is another Group 2 hydroxide.

Solutions of calcium hydroxide and barium hydroxide are both alkalis and turn pH paper purple.

State how you could distinguish between calcium hydroxide and barium hydroxide using a flame test. [1]

.....

.....

